

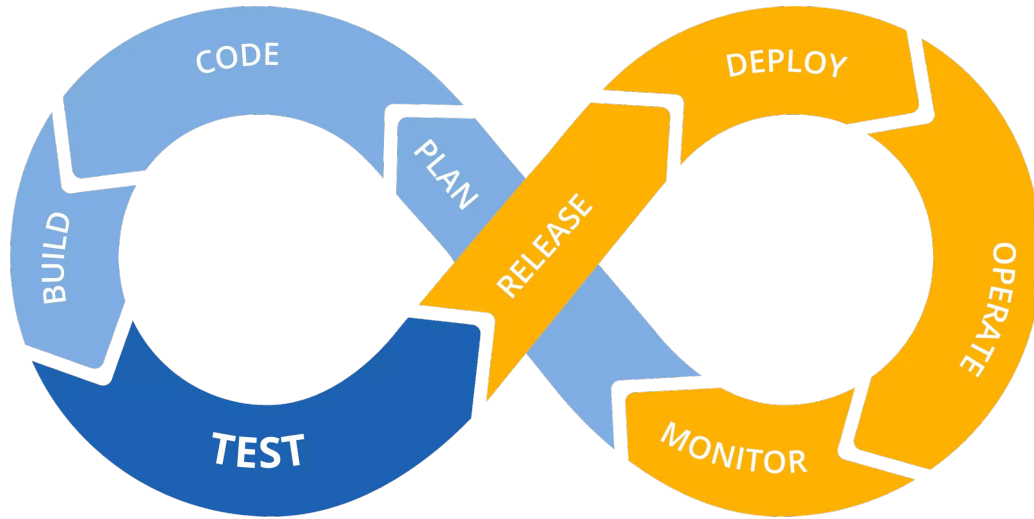
AWS CI/CD Pipeline



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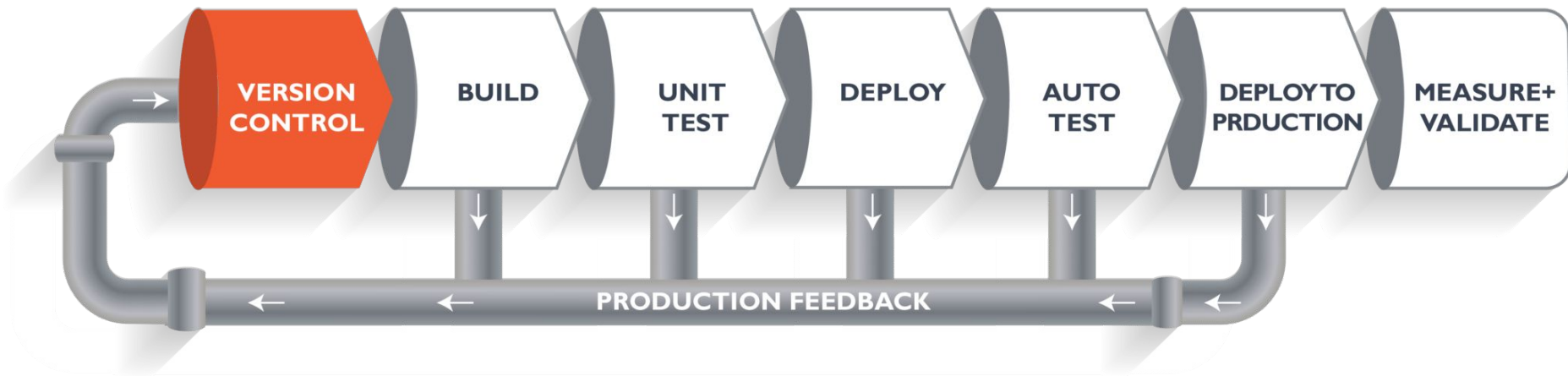
What is DevOps?



DevOps is a software development approach which involves:

- continuous development,
- continuous testing,
- continuous integration,
- continuous deployment
- continuous monitoring of the software throughout its development lifecycle

CI/CD





Version Control (Code)



Build



Test

Deploy



CI/CD Pipeline



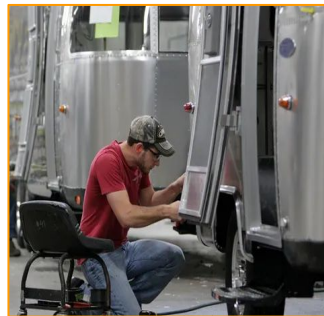
Version Control (Code)



Build



Test



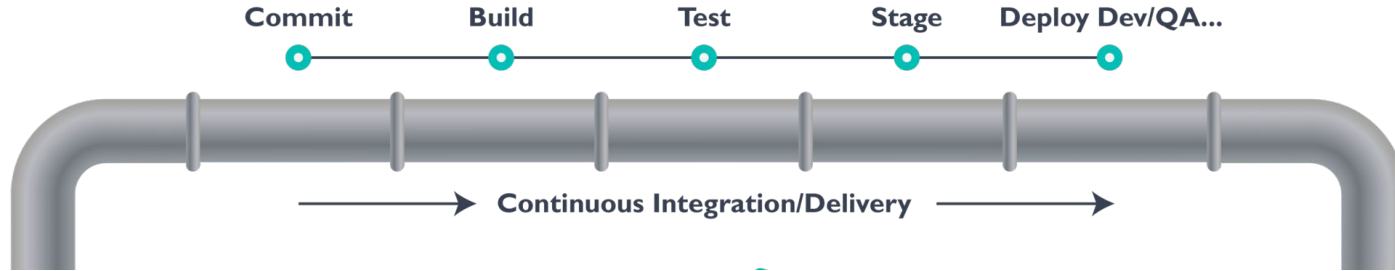
Deploy



3

CI/CD Pipeline Stages

CI/CD Pipeline



Gitlab



CodePipeline

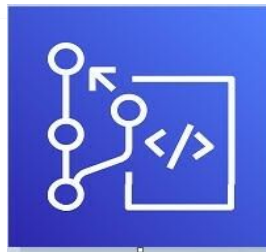
Teamcity



Version Control Stage



Bitbucket
Atlassian

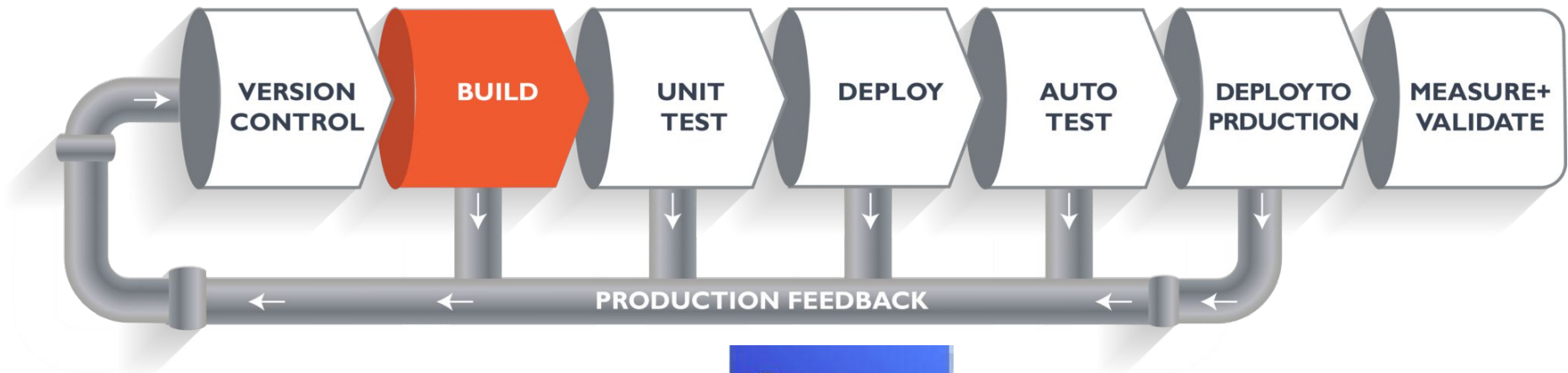


CodeCommit

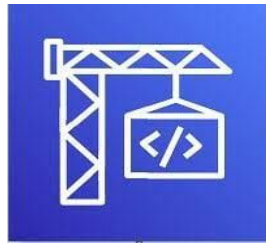


Gitlab
Gitlab

Build Stage



Maven™



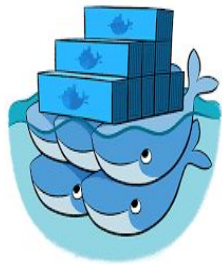
CodeBuild



Gradle



Deploy Stage

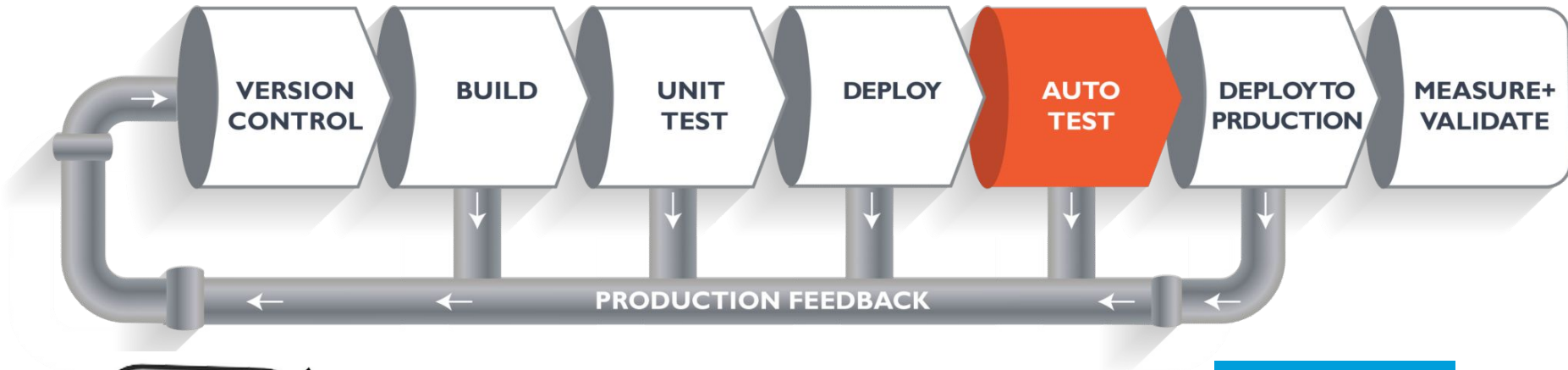


CodeDeploy

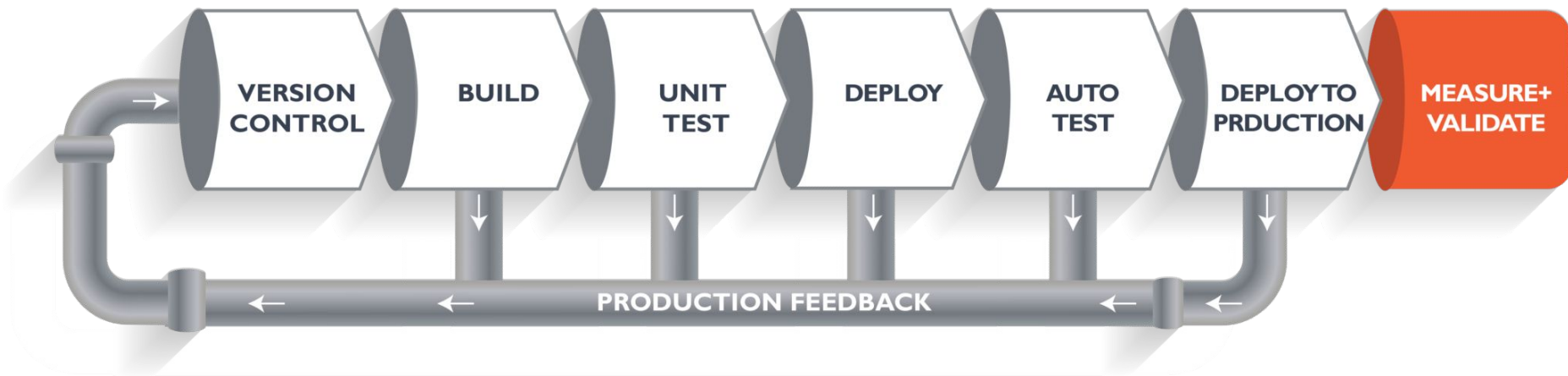


Auto Test Stage

User Acceptance Tests
Functional Tests
Load Tests
Security Tests



Measure + Validate



Nagios[®]



Continuous Delivery vs Continuous Deployment

Continuous Integration



Continuous Delivery



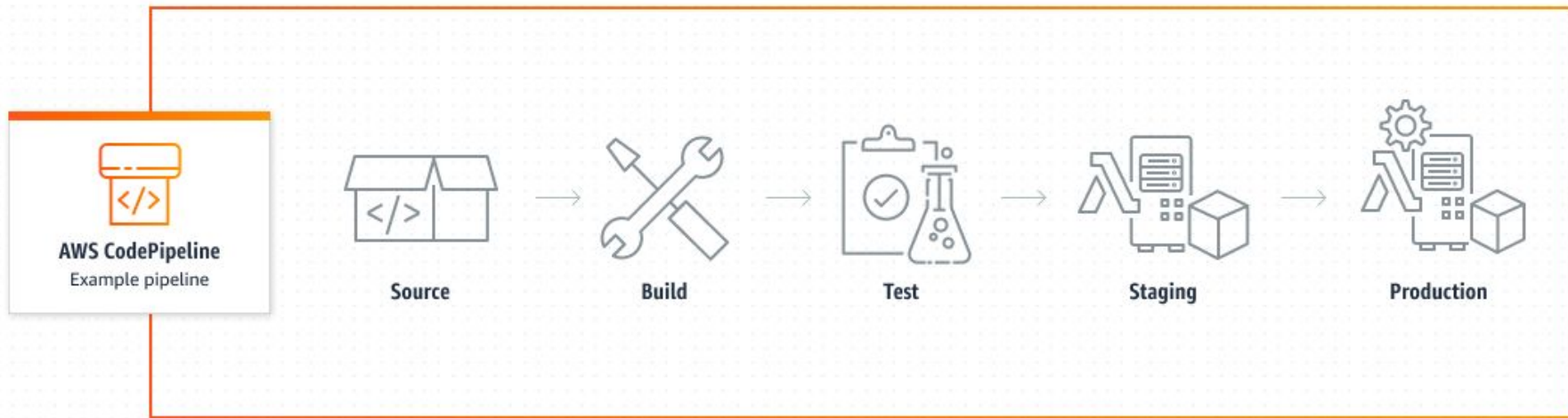
Continuous Deployment





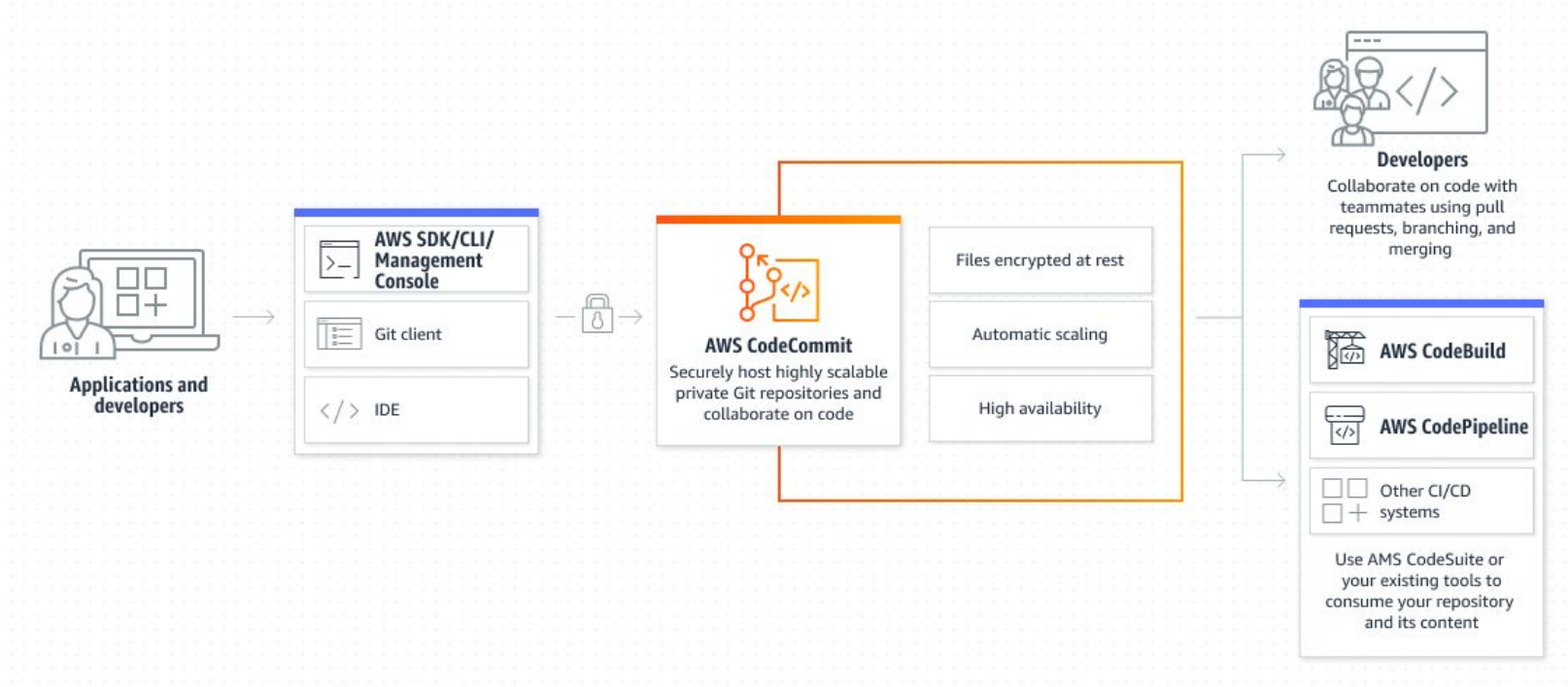
CI/CD Pipeline with AWS

AWS CodePipeline



AWS CodePipeline is a continuous delivery service you can use to **model, visualize, and automate the steps required to release your software.**

AWS CodeCommit

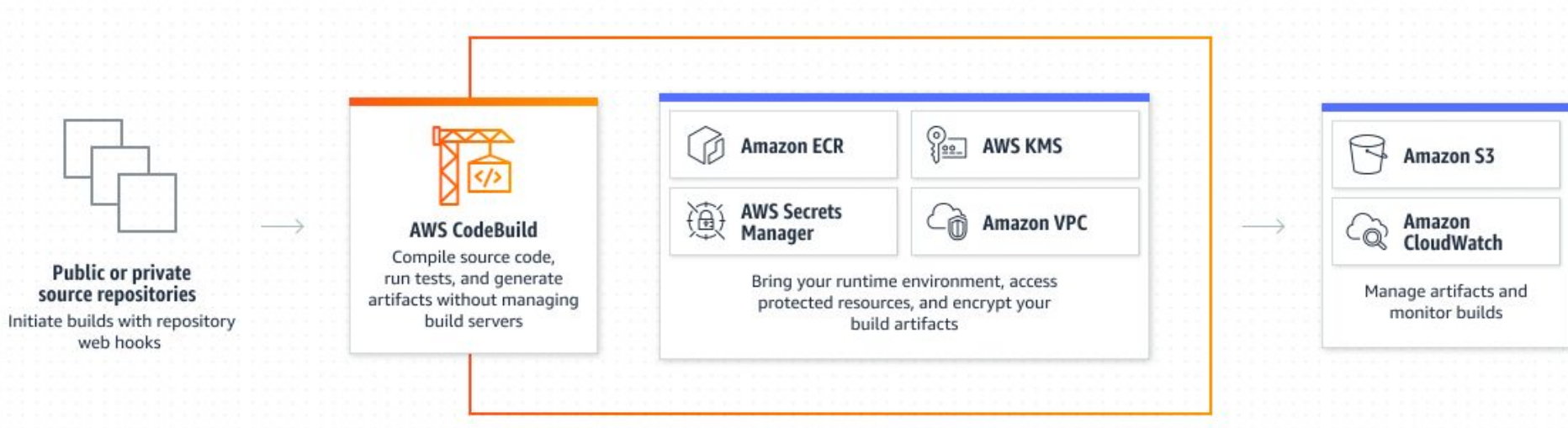


AWS CodeCommit is a **version control service** hosted by Amazon Web Services that you can use to **privately** store and manage assets (such as documents, source code, and binary files) in the cloud.

AWS CodeCommit

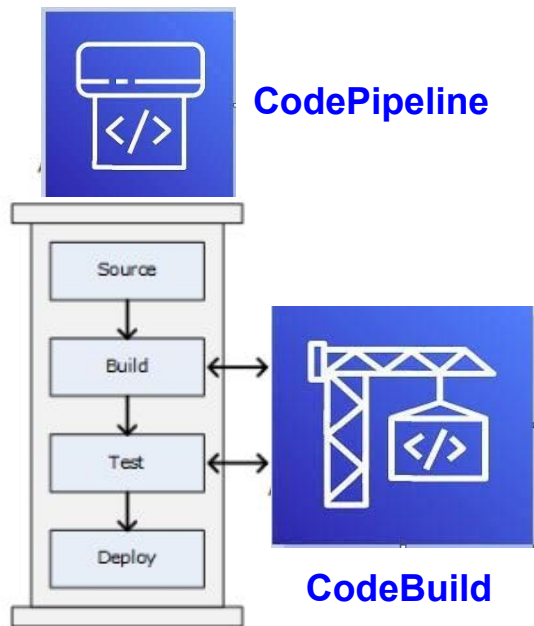
- It supports the standard functionality of Git, so **it works seamlessly** with your **existing Git-based tools**.
- It hosts **private repos**
- CodeCommit repositories are **encrypted at rest as well as in transit**.
- **Support pull requests** where users can review and comment on each other's code changes before merging them to branches
- CodeCommit has **no limit on the size** of your repositories
- You can **migrate to CodeCommit from any Git-based repository**.

AWS CodeBuild



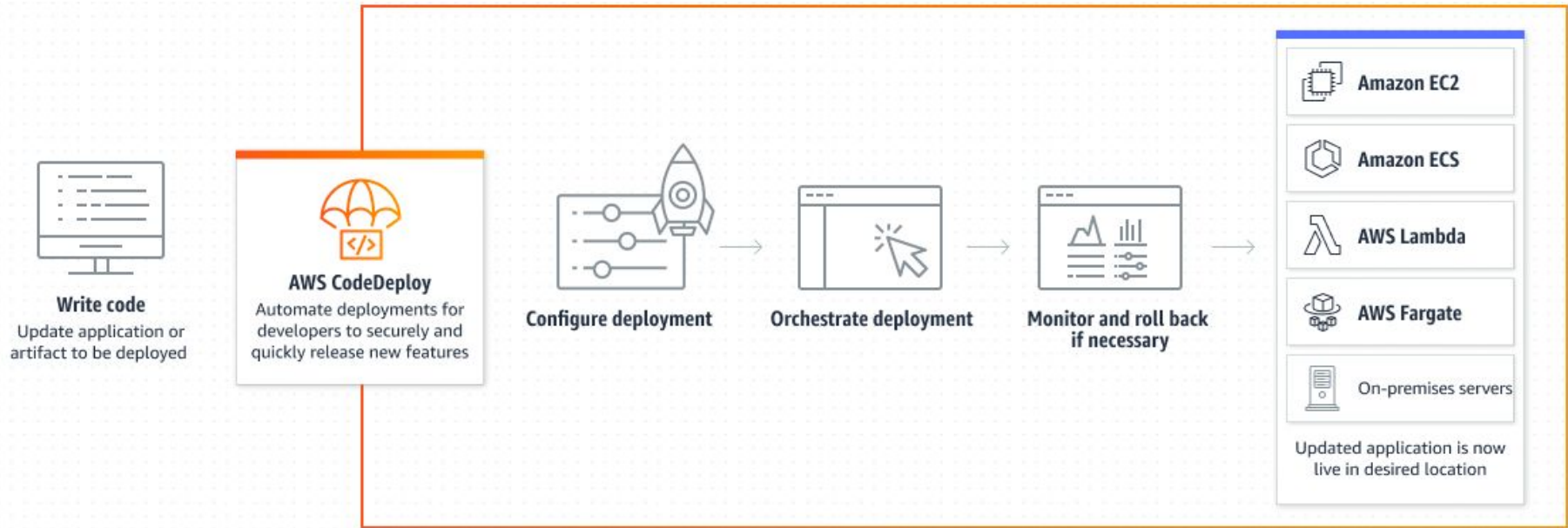
- **compiles** your source code,
- runs **unit tests**,
- **produces artifacts** that are ready to deploy.

AWS CodeBuild



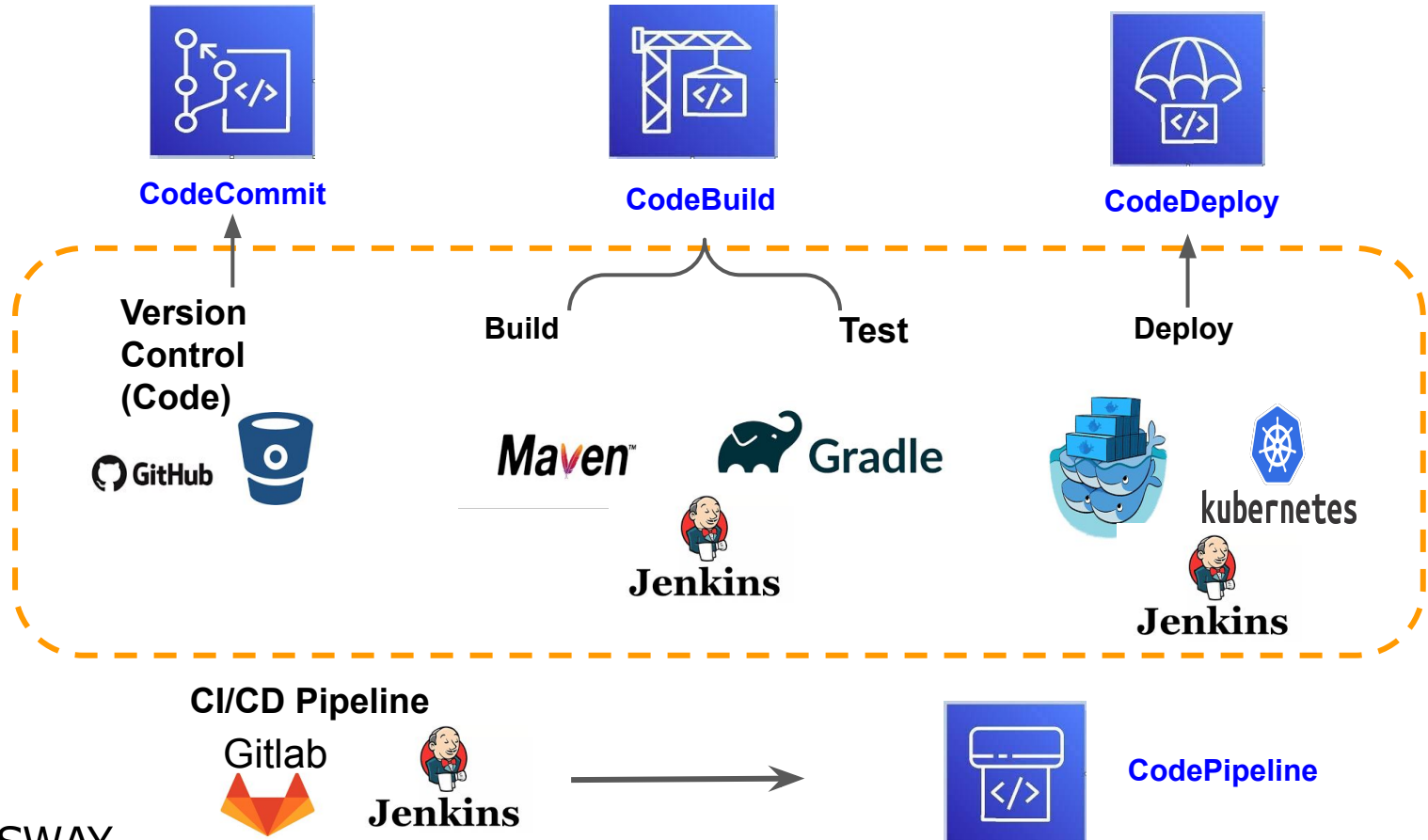
- **Fully managed** – CodeBuild eliminates the need to set up, patch, update, and manage your own build servers.
- **On demand** – CodeBuild scales on demand to meet your build needs. **You pay only for the number of build minutes you consume.**
- **Out of the box** – CodeBuild provides preconfigured build environments for the most popular programming languages. All you need to do is point to your build script to start your first build.
- **Easily reachable** from Codepipeline Console

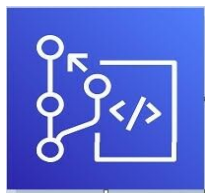
AWS Deploy



AWS CodeDeploy is a fully managed deployment service that automates software deployments to various compute services, such as **Amazon Elastic Compute Cloud (EC2), **Amazon Elastic Container Service (ECS)**, **AWS Lambda**, and **your on-premises servers**.**

AWS Pipeline Structure





CodeCommit



CodeBuild



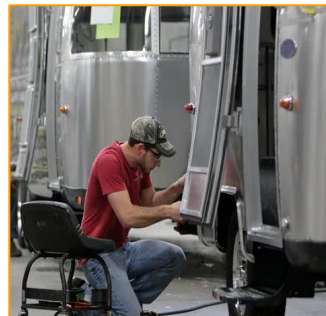
CodeDeploy

**Version Control
(Code)**

Build

Test

Deploy



CodePipeline



CI/CD Pipeline

